EDA-Graph: Graph Signal Processing of Electrodermal Activity for Emotional States Detection

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***Abstract*—The continuous detection of emotional states has many applications in mental health, marketing, human-computer interaction, and assistive robotics. Electrodermal activity (EDA), a signal modulated by sympathetic nervous system activity, provides continuous insight into emotional states. However, EDA possesses intricate nonstationary and nonlinear characteristics, making the extraction of emotion-relevant information challenging. We propose a novel graph signal processing (GSP) approach to model EDA signals as graphical networks, termed EDA-graph. The GSP leverages graph theory concepts to capture complex relationships in time-series data. To test the usefulness of EDA-graphs to detect emotions, we processed EDA recordings from the CASE emotion dataset using GSP by quantizing and linking values based on the Euclidean distance between the nearest neighbors. From these EDA-graphs, we computed the features of graph analysis, including total load centrality (TLC), total harmonic centrality (THC), number of cliques (GNC), diameter, and graph radius, and compared those features with features obtained using traditional EDA processing techniques. EDA-graph features encompassing TLC, THC, GNC, diameter, and radius demonstrated significant differences ($p < 0.05$) between five emotional states (Neutral, Amused, Bored, Relaxed, and Scared). Using machine learning models for classifying emotional states evaluated using leave-one-subject-out cross-validation, we achieved a five-class F1 score of up to 0.68.**

***Index Terms*—Emotional states, electrodermal activity, graph signal processing.**

List of Abbreviations

EDA:	Electrodermal Activity, the physiological signal analyzed in this study.
GSP:	Graph Signal Processing, the core analytical methodology applied to EDA signals.
uS:	Microsiemens, unit of measurement for conductivity.
SCR:	Skin Conductance Response.
SCL:	Skin Conductance Level.
nsSCR:	non-specific Skin Conductance Response.
TVSymp:	Time-Varying Spectral EDA index.
EDASymp:	Low-frequency Sympathetic component of EDA.
$x_{original}$:	Original continuous EDA signal.
$x_{quantized}$:	Quantized (discretized) EDA signal.
X_{nodes}:	Set of unique quantized EDA values (nodes).
$X_{quantized_{shifted}}$:	Quantized signal shifted by one sample.
Q:	Quantization step size, used in discretizing EDA signals into quantized values.
N:	The set of nodes in the EDA-graph, representing unique quantized values of EDA signals.
E:	The set of edges in the EDA-graph, representing connections between nodes based on Euclidean distance.
D_{ij}:	Euclidean distance between nodes i and j in the EDA-graph.
M:	Dimensionality of the signal (in this case, 1 for univariate EDA).
x_{ik},x_{jk}:	k_{th} dimension of nodes x_i and x_j.
D:	Distance matrix of size $N\ x\ N$, where N is the number of nodes.
x_s,x_t:	Source and target nodes.
w_{st}:	Weight (strength) of the connection between x_s and x_t.
A:	Adjacency matrix of the EDA-graph, representing the connectivity between nodes.
a_{ij}:	Element of the adjacency matrix A, representing the weight of the connection between nodes x_i and x_j.
TLC:	Total Load Centrality, a graph metric indicating the importance of nodes based on network flow.
THC:	Total Harmonic Centrality, a measure of node centrality based on harmonic mean distances.
GNC:	Graph Number of Cliques, indicating the number of fully interconnected subgraphs.
PAA:	Piecewise aggregate approximation.
F1:	F1 scores, a metric used to evaluate the performance of machine learning classification.
GNB:	Gaussian Naive Bayes.

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KNN:	K-Nearest Neighbors.
RF:	Random Forest.
BEC:	Bagging Classifier.
GBC:	Gradient Boosting Classifier.
DT:	Decision Tree.
SVM:	Support Vector Machine.
GNN:	Graph Neural Network.

I. Introduction

UNDERSTANDING emotional states is a complex but crucial task, given their significant impact on various aspects of life and technology [1]. Emotional states, encompassing subjective experiences, cognitive processes, physiological responses, and behavioral expressions, play a pivotal role in fields such as mental healthcare, human-computer interaction, and assistive robotics [2], [3]. For instance, accurately identifying emotional states like anxiety and sadness is key in treating mood disorders, which affect a substantial global population [4], [5]. In the realm of human-computer interactions, recognizing and responding to emotional states such as confusion or frustration can greatly enhance user experience [6]. Similarly, assistive robots that can perceive and adapt to human emotional states promise more seamless human-robot collaboration [2], [6].

Emotional states recognition employs physiological modalities, such as electrodermal activity (EDA), Electroencephalography EEG, and electromyography, which directly measure internal autonomic activity [7]. This enables continuous, covert insights unaffected by display rules. In contrast, non-physiological methods like facial expressions and speech processing analyze observable behaviors that can be intentionally masked [8], [9]. Physiological emotional states recognition enables continuous, covert insights unaffected by display rules, as it directly measures the internal autonomic nervous system activity [10], [11].

Among physiological signals, EDA is attractive because of its sensitivity to psychological arousal and reactivity modulated by the sympathetic nervous system [12], [13]. EDA comprises tonic and phasic components. EDA refers to changes in the electrical conductivity of the skin caused by sweat gland activity, which is modulated by the sympathetic nervous system [14]. EDA comprises tonic and phasic components [15]. The tonic component of EDA reflects a slowly varying baseline conductivity influenced by circadian rhythms, hormones, thermoregulation, and hydration [16]. In contrast, the phasic component of EDA represents rapid bursts linked to stimulus processing and cognitive load. The phasic component of the EDA contains waves called skin conductance responses (SCRs) [15]. The morphology, amplitude, and timing of SCRs provide insight into emotional and cognitive states [12]. Analysis of tonic and phasic EDA can reveal information about psychological arousal and reactivity modulated by the sympathetic nervous system [17]. However, long-term habituation effects and non-stationarity pose further challenges [16]. Thus, extracting reliable information from a nuanced EDA waveform remains challenging [12].

Various techniques have been applied to analyze the nonlinear and nonstationary characteristics of EDA, including time-domain, frequency-based, time-frequency, and nonlinear methods [18]. However, these approaches have limitations in disentangling the intricate emotional states-signal relationships in EDA because of their complex nonstationary and nonlinear nature [19]. Time-domain analysis extracts features such as the mean, variance, skewness, and kurtosis to quantify overall trends and variations [20]. However, time-domain techniques cannot assess abrupt or nonstationary variations and lack frequency information that can provide insights into sympathetic dynamics [21]. Overall, these cases exemplify not only how each method has specific utilities, but also areas for improvement in fully unraveling emotional states-signal relationships in EDA data.

Advanced signal processing techniques, such as Fourier transform and wavelet transform, extract features, such as power spectral density, to characterize frequency bands [22]. Time-frequency techniques such as the short-time Fourier transform provide combine d time-frequency representations, but they have reduced resolution due to the inherent time-frequency trade-off [23]. Variable-frequency complex demodulation offers improved resolution for EDA analysis [21]. These studies have shown promising results; however, they seem unable to fully unravel intricate emotional states-signal relationships because classification accuracy using these features is relatively low. More sensitive approaches are required to model dynamics beyond individual time, frequency, or nonlinear features.

Recent studies propose innovative methods to address gaps in emotion recognition, leveraging advanced signal processing and deep learning techniques. Raheel et al. (2020) utilized pre-processing with a Savitzky-Golay smoothing filter to enhance the precision of physiological signals, including EEG, photoplethysmography, and EDA, for emotion classification, achieving a classification accuracy of 79.76% for four emotions using the fusion of EEG, EDA, and photoplethysmography features [24]. Pan et al. (2022) employed a two-stage feature optimization method based on feature correlation analysis and ReliefF algorithm to reduce feature dimensionality and improve emotion recognition accuracy, achieving 98.40% accuracy on the AuBT dataset and 92.34% on the DEAP dataset [25].

Deep learning techniques have significantly advanced emotion recognition from physiological signals in recent years. Zhang et al. (2023) introduced an Emotion recognition algorithm based on Deep Siamese Networks for fine-grained valence and arousal classification, achieving averaged accuracies of 76.04%, 76.62%, and 57.62% for one-dimension binary valence, arousal, and two-dimensional 5-class classification, respectively, using only 5 training samples per class [26]. Singh et al. (2023) explored data augmentation and channel selection techniques on the DEAP dataset, obtaining 92.5% and 81.25% accuracy for two-class classification in valence and arousal emotions [27]. Ganapathy et al. (2021) proposed a convolutional neural network-based approach for learning robust features from EDA signals, outperforming most state-of-the-art methods with an F1-scores (F1) of 79.30% and 71.41% for arousal and valence dimensions [28]. Govarthan et al. (2023) developed a deep learning framework using a configurable convolutional neural network for categorical emotional state assessment, achieving an average classification accuracy of 80.20%, recall of 60.41%, specificity of 86.8%, precision of 60.05%, and F1 of 58.61% [29]. Pinzon-Arenas et al. (2023) explored the application of a parallel hybrid architecture called the parallel Temporal Convolutional Network Self-Attentive Bottom-Up Long Short-Term Memory for continuous emotion detection, demonstrating its suitability with an average root mean square error of 1.585 across different scenarios [30].

Graph Signal Processing (GSP) offers a novel solution to these challenges. Utilizing concepts from graph theory, GSP represents time-series data like EDA as graphs, capturing intricate relationships within the data. This approach allows for a more

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nuanced analysis, potentially unveiling patterns and connections that conventional methods might miss. In this study, we pioneer the application of GSP to EDA data for emotional states detection, conceptualizing EDA signals as graph networks, referred to as EDA-graph.

GSP provides a framework for extending traditional time-series signal processing techniques to data with complex relationships that are representable as graphs [31]. GSP represents time-series data as graphs with nodes as signal values and edges based on similarities, allowing the application of graph theory concepts [31]. The graph representation encodes intrinsic connectivity patterns and complex relationships lost in simplistic temporal processing approaches. Other mathematical concepts have been used with GSP to facilitate the analysis of complex data, including graph Fourier transform, graph filtering, and spectral graph theory [29]. These techniques enable the transformation of time-series data into graphical representations that encode critical dynamics and then analyze the graph signals using filtering and spectral principles to characterize key components and relationships. Recently, GSP has been successfully applied in processing brain networks, social networks, point clouds, and other irregular data [32].

This work presents a pioneering application of GSP for analyzing EDA signals to classify emotional states. The novelty of our work lies in the systematic evaluation and adaptation of graph representations tailored for emotional state detection from EDA signals. This novel GSP approach addresses a significant gap by customizing the signal representation to capture the unique morphological characteristics of EDA waveforms. We introduce innovative graph metrics explicitly designed to encode the physiological underpinnings of EDA and its association with different emotional states. By leveraging these tailored graph representations, our approach enables the detection of subtle patterns linked to nuanced emotional experiences, successfully extracting discriminative features that uncover distinctions between emotional states often overlooked by traditional EDA analysis techniques.

Key contributions include:

- The introduction of a novel graph-based framework for EDA signal analysis, marking a significant advancement in the field of emotional state detection. This framework is specifically designed to accommodate the unique characteristics of EDA signals, offering a more nuanced understanding of emotional states.
- The development and application of custom graph metrics tailored to EDA signals, enabling the identification and analysis of subtle emotional patterns. These metrics enhance the sensitivity of emotional state detection, demonstrating the potential of graph-based approaches in psychophysiological research.
- The implementation of a rigorous comparative analysis, which clearly establishes the superiority of our graph-based method over existing EDA analysis techniques, including advanced non-linear signal processing methods such as isaxEDA, comEDA, topEDA, and netEDA. This validation underscores the effectiveness and precision of our approach in classifying emotional states.
- Introduction of a comprehensive feature extraction strategy that incorporates dynamics, complexity, structure, and network properties of EDA signals. Coupled with an innovative segmentation method, our approach ensures precise emotional state analysis, setting a new standard for accuracy in EDA-based emotion recognition.

TABLE I
CASE DATASET DESCRIPTIONS

Feature	Data	Description
Participants	30 (15 males, 15 females)	Age range 18-40 years (mean: 26.5 years); Right-handed normal or corrected vision; No history of neurological disorders
Age	18 to 40 years old	Mean: 26.5 years
Video clips Stimuli	8	Clips conveying emotions - Amused, Relaxed, Bored, Scared, and Neutral videos. ~2 minutes.
Video clips non-stimuli	3	Start video, blue screen separating videos with stimuli, and the completion video which is also blue screen. The videos are ~2 minutes.
Participant ratings	Arousal and Valence	Participants rated emotional Arousal and Valence continuously in a scale from 0.5 to 9.5
EDA Recording	Continuous EDA data during the video stimulation	Technology ProComp Infiniti encoder Skin Conductance Sensor, SA9309M, sampled at 1 kHz.

II. MATERIALS AND METHODS

A. Dataset and Emotional States Labeling

For this study, EDA signals were obtained from the "Continuously Annotated Signals of Emotion (CASE)" database, which is publicly available [10]. The database consists of physiological modalities including electrocardiogram, blood volume pressure, electromyography, and EDA obtained from 30 subjects while watching eight video clips aimed at eliciting different target emotional states.

The video stimuli were approximately 2–3 minutes in length and aimed to elicit different target emotions. The order of video clips was varied in a pseudo-random fashion for each participant to avoid carry-over effects from previous videos, resulting in a unique video sequence for every participant. A two-minute blue screen separated each video, allowing participants to provide continuous annotation of their emotional state by rating valence (positive/negative) and arousal (calm/excited) on a scale from 0.5 to 9.5, as described in Table I. To align our study with a more structured emotional framework, we adopted discrete emotional state labels derived from specific intervals of arousal and valence scores. This re-labeling yielded five distinct categories: Neutral, Amused, Bored, Relaxed, and Scared. Our approach is grounded in the circumplex model of emotions, particularly focusing on the reduction of ambiguity around the central (neutral) region of the model. We determined 'Neutral' based on a specific range within the arousal and valence scales from 4.5 to 5.5, thus creating a clear demarcation for this often-ambiguous emotional state, as shown in Fig. 1 that detail the labeling criteria and

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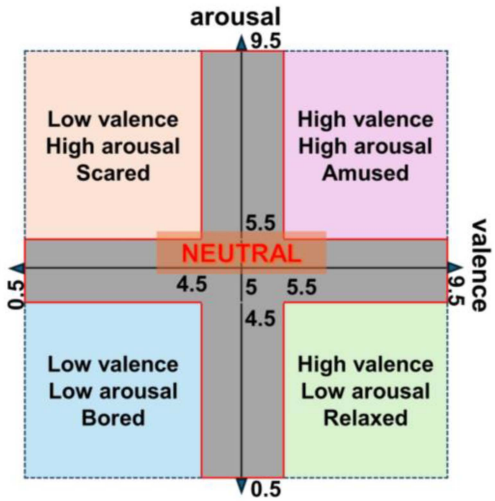

Fig. 1. Representation of emotion states in the plane, showing the ranges of arousal and valence that represent the emotional states: relaxed: high valence-low arousal, amused: high valence-high arousal, bored: low valence-low arousal, scared: low valence-high arousal, and neutral that is a cross-divider.

the corresponding emotional states as derived from the specific ranges in arousal and valence.

Kuppens et al. (2022) [33] emphasized the importance of including the neutral state in emotional process analyses, which informed our decision to incorporate it in our study. This understanding informed our methodology and aimed to broaden the scope of our investigation into emotional states. Building on this foundation, we sought direct comparability with the pioneering work of Zhang et al. (2023) on few-shot learning for emotion recognition, which also employed the CASE dataset and defined neutral as valence and arousal ratings within the range of 4.5 to 5.5 on a 9-point scale [26]. This alignment allows for a nuanced comparison and contributes to the consistency and relevance of our research within the existing literature.

This decision to transition from continuous to discrete emotional labeling is driven by our study's objective: to rigorously evaluate the effectiveness of the EDA-Graph approach in emotional states detection. Our discrete labeling does not reflect the intended emotions of the video stimuli in the CASE dataset but is solely based on the participants' arousal and valence scores. This ensures that our analysis accurately represents the emotions experienced by the subjects, not just the emotions intended to be elicited by the videos.

B. Preprocessing of EDA Signals

In the CASE dataset, EDA signals were acquired using a skin conductance sensor (SA9309M, ProComp Plus, USA) at a sampling rate of 1kHz. We downsampled the EDA signals to a frequency of 8 Hz using standard decimation. This downsampling retains adequate samples to characterize the EDA dynamics [31], while significantly reducing the data volume for faster processing. Following downsampling, the signals were passed through a fourth-order low-pass Butterworth filter with a cut-off frequency of 1 Hz. The 1 Hz cut-off was selected to eliminate unnecessary components while retaining the relevant phasic drivers and tonic baseline shifts that occur below this threshold. Additionally, a median filter (1-second window, 8 samples) was applied to further enhance the quality of the EDA signals prior to analysis [9].

C. Obtaining EDA-Graphs

In GSP, the selection of nodes and edges is crucial as it directly influences the interpretation of physiological data. Nodes in our EDA graphs represent specific signal values, encapsulating key information about physiological states. The edges, defined based on the similarities between these values, characterize the smoothness or abruptness of EDA changes. This setup allows us to interpret stronger connections between successive nodes as gradual EDA variations, commonly seen in steady emotional states, while weaker connections might indicate sharp shifts, potentially linked to sudden emotional disruptions or excitations.

To obtain EDA-graphs using GSP, the preprocessed EDA signals were segmented into 60-second windows with a 50% overlap between consecutive windows. For each window, we checked whether at least 80% of the samples were consistently labeled with the same emotional state (Neutral, Amused, Bored, Relaxed, and Scared) based on the dataset's continuous annotation of arousal and valence. This criterion ensured that the analyzed windows were largely related to a single emotional state. Only windows that met this criterion were included in further analysis.

The resulting dataset consisted of 16, 447 windows, distributed among different emotional classes as follows: Neutral: 12, 513 samples, Amused: 825 samples, Bored: 413 samples, Relaxed:1, 807 samples, and Scared: 889 samples. The step-by-step process of generating an EDA-graph is as follows:

1) Step 1: Quantization: Quantization (Q) involves discretizing a continuous EDA signal x_{original} into discrete values $x_{\text{quantized}}$. The Q process can be mathematically expressed as:

$$x_{\text{quantized}} = Q \cdot \text{round}\left(\frac{x_{\text{original}}}{Q}\right) \tag{1}$$

The Q step size controlled the resolution of the quantized signal as expressed in (1). By dividing the original signal's value range into non-overlapping intervals of size Q, the continuous amplitude can be discretized into a finite set of values.

The choice of the optimal Q value is crucial for accurately capturing the subtle variations in EDA signals, which are essential for understanding emotional states. To determine the most effective Q value, we conducted a systematic analysis by evaluating the performance of our graph-based approach across a wide range of Q levels, from 0.001 to 0.99 microSiemens (μS). Based on our detailed comparative analysis, a value of $Q = 0.05$ μS emerged as the optimal choice.

The selection of $Q = 0.05$ µS aligns with the typical value used to define significant changes in EDA signals, as suggested by previous literature [12]. In Step 1 of Fig. 2, we illustrate a comparison between the original and quantized segments, showcasing the effect of the Q process on the EDA signal.

2) Step 2: Node Definition: Nodes in a graph are fundamental units that represent data points or entities. In the context of quantized EDA signals, nodes correspond to the unique values of the quantized signal within a segment. For example, if a quantized EDA signal contains five unique values, these values form the nodes of the graph, capturing the discrete dynamics of the signal (Cai, Zheng, & Chang, 2017) [34].

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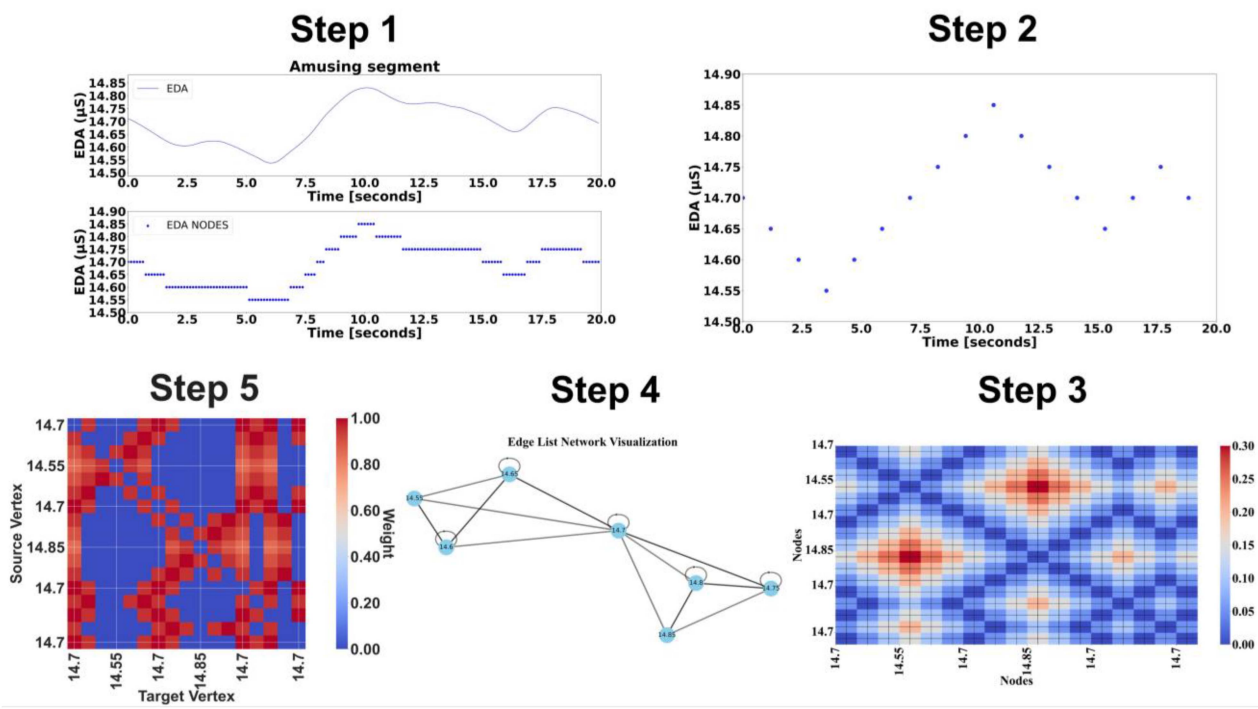

Fig. 2. Steps for EDA-graph generation and its representation using adjacency matrix. Step 1: preprocess the EDA signal, excluding any repeated values resulting from Q. Step 2: only the first instance of each unique value as a node in the graph. Step 3: the euclidean distance between each pair of nodes is calculated. Step 4: connect each node to its k-nearest neighbors. Step 5: generate the adjacency matrix A, which uses the inverted weights of the distance; the nearest values have a value of 1.

Edges in a graph represent the relationships or connections between the nodes. For quantized EDA signals, edges can be defined through methods like Euclidean distance calculation between nodes, capturing the similarity or dissimilarity in signal values. Connections might be limited to a predefined number of nearest nodes, such as the eight nearest in this context, to reflect intrinsic connectivity patterns in the EDA data efficiently (Liu & Tang, 2021) [35].

After quantizing the EDA segment, the next step is to set up nodes. In our approach, the nodes represent the unique values of the quantized EDA signal, that is, the unique values that appear in $x_{\text{quantized}}$ in the given segment. For example, if the quantized EDA signal has five unique values, $\{0.5,\ 1.0,\ 1.5,\ 2.0,\ 2.5\}$, this set of five quantized values becomes the graph nodes, $\mathrm{N} = \{0.5,\ 1.0,\ 1.5,\ 2.0,\ 2.5\}$, as shown in step 4 of Fig. 2. All the samples in the segment with the same value are represented by the same node. This can be expressed as in (2):

$$X_{\text{nodes}} = X_{\text{quantized}}\ \left[X_{\text{quantized}} \neq X_{\text{quantized}_{\text{shifted}}}\right] \tag{2}$$

where $X_{\text{quantized}_{\text{shifted}}}$ is the quantized signal shifted by one sample. This node definition directly captures the discrete dynamics of the $X_{quantized}$ EDA signal within the graph structure. The number of nodes depends on the resolution of the $X_{quantized}$ and the variability of EDA levels observed during the recording. Finer Q resolution produces more potential quantized values and graph nodes. $x_i = \{x_1, x_2, \ldots, x_{N_q}\}$, where $N_q$is the total number of unique $X_{quantized}$ values present in the signal.

3) Step 3: Distance Calculation: The calculation of pairwise distances between nodes is a crucial step in constructing the graph representation of the EDA signal. While several distance metrics can be employed, such as Euclidean, Cosine, and Correlation distances, our exploration revealed that the Euclidean distance approach yielded the most discriminative power in capturing the nuances of emotional states.

The Euclidean distance (D_{ij}) between two nodes, x_i and x_j is defined as:

$$D_{ij} = \sqrt{\sum\nolimits_{k=1}^{M} (x_{ik} - x_{jk})^2} \tag{3}$$

Here, M represents the dimensionality of the signal, which in our case is 1. We consider the quantized EDA signal as a one-dimensional time series, as expressed in (3). The variables x_{ik} and x_{jk} correspond to the k_{th} dimensions of nodes x_i and x_j. In our case, D_{ij} is the difference in value between each pair of nodes.

The calculated distances are organized into a matrix D of size $N \times N$, where N is the number of nodes. This distance matrix encapsulates the pairwise dissimilarities between all nodes, serving as the foundation for constructing the graph structure.

The Euclidean distance metric was chosen after extensive evaluation, as it outperformed other distances like Cosine and Correlation in capturing subtle EDA signal variations and discriminating emotional states effectively. Its sensitivity to magnitude and directional changes in the signal allowed accurate quantification of dissimilarities between segments, enabling informative graph representations that preserved the intricate patterns and nuances associated with different emotional experiences in the underlying physiological signal.

4) Step 4: Nearest Nodes Connection: Based on empirical exploration, we restricted connections between nodes to the 8 nearest nodes. For each x_i, we select its closest 8 nodes based on

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distance D_{ij}. This step connects x_i and x_j that are most similar to each other and forms the edge set E_s This can be represented as follows below in (4). The choice of 8 nearest neighbors is based on the sampling frequency of the EDA signal, which was 8 Hz in this study. Connecting each node to its 8 closest nodes based on the sampling rate captures the intrinsic connectivity patterns in the EDA data.

$$E_s = \{(x_s, x_t, w_{st})\} \tag{4}$$

x_s and x_t represent the source and target nodes, respectively. The variable w_{st} represents the strength or weight of the connection between x_s and x_t, typically determined by the distance value from D. We defined w_{st} as inversely proportional to the distance between x_s and x_t as represented in (5).

$$w_{st} = \frac{1}{D_{st}} \tag{5}$$

Connecting each node x_i to its 8 closest x_j results in a comprehensive representation of the graph's edges. Mathematically, this can be expressed as follows:

$$E = \bigcup_{i=1}^{8} \{(x_i, x_j, w_{ij})\} \tag{6}$$

This process is intended to allow the graph to capture the underlying relationships between signal values. The concept is illustrated in Fig. 2. After this step, the EDA segment is represented using E, which encapsulates the inherent relationships within the physiological responses, ultimately providing insights into the dynamics of emotional states. The choice to restrict connections between nodes to the 8 nearest neighbors was empirically found to match the sampling frequency of the EDA signals. With a sampling rate of 8 Hz in this study, connecting each node to its 8 nearest nodes based on distance effectively captures the intrinsic connectivity patterns within the EDA data at that resolution.

Although 8 nearest neighbors were suitable for the 8 Hz sampling rate used here, it is important to note that this parameter is dependent on the sampling frequency and not universally fixed at 8. In general, the number of nearest node connections can be matched to the sampling rate of the EDA signal being analyzed. This establishes node connections at intervals corresponding to the data resolution. The relationship between sampling rate and nearest neighbor connections was determined through empirical exploration of graph architectures on sample EDA signals. Comparing graphs built using different numbers of nearest neighbors revealed that linking nodes based on the sampling frequency optimized representation of the EDA data relationships and features.

While the current study used 8 based on the 8 Hz sampling rate, the same principle would apply at other frequencies. For example, with a sampling rate of 10 Hz, connecting each node to its 10 nearest nodes would capture the intrinsic patterns. This adaptive approach allows the EDA graph architecture to flexibly match varied sampling rates.

5) Step 5: Adjacency Matrix: Following the establishment of the strength of the connections between nodes, we proceeded to construct the adjacency matrix, A, that represents the EDA-graph. A is a square matrix with dimensions $N \times N$, in which each element a_{ij} signifies the weight of the connection between nodes x_i and x_j. In this context, a_{ij} is defined in (7).

$$A_{ij} = \begin{cases} w_{ij}, & \text{if } (x_i, x_j, w_{ij}) \in E \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

A serves as a concise representation of the graph's connectivity, enabling efficient representation of the network's structure.

TABLE II
GRAPH-LEVEL, NODE-LEVEL AND EDGE-LEVEL FEATURES OBTAINED FROM EDA-GRAPHS

Level	Graph Features
Graph	Total Triangle Number, Graph Energy, Transitivity, Cliques Counts, Number of Cliques, Graph Is Chordal, Center, Diameter, Radius, Periphery, Average Clustering, Analyze Weisfeiler-Lehman Kernel, Graph Spectrum (GS) Std, Laplacian Spectrum (LS) Std, Avg Triangle Participation, GS Mean Magnitude, GS Min Magnitude, GS Max Magnitude, GS Median Magnitude, GS Skewness Magnitude, GS Kurtosis Magnitude, LS Mean Magnitude, LS Min Magnitude, LS Max Magnitude, LS Median Magnitude, LS Skewness Magnitude, LS Kurtosis Magnitude, GS Mean Phase, GS Min Phase, GS Max Phase, GS Median Phase, GS Skewness Phase, GS Kurtosis Phase, LS Mean Phase, LS Min Phase, LS Max Phase, LS Median Phase, LS Skewness Phase, LS Kurtosis Phase [36]
Node	Total Degree Centrality, Total Closeness Centrality, Total Betweenness Centrality, Total Eigenvector Centrality, Total Load Centrality, Total Harmonic Centrality, Total Pagerank, Total Hubs, Number Of Nodes, Maximum Degre, Minimum Degre, Mediam Degree, Closeness Centrality, Eccentricity [37]
Edge	Total Flow Centrality, Total Log Flow Centrality, Number of Edges, Assortative, Spearman Correlation [38].

D. Feature Extraction

1) EDA-Graph Features: Features were obtained from the EDA-graph of each segment at the graph-, node-, and edge-levels, to fully capture the characteristics of the topology of the EDA-graph. Node-level features represent properties of individual nodes in the graph, and statistical measures (mean, standard deviation, maximum, etc.) across nodes were computed as features of the EDA-graph. Edge-level features encoded relationships and connectivity between node pairs. Node- and edge-level features describe interaction patterns and signal transitions [39]. The graph-level features reflect holistic connectivity, complexity, and emotional states-related organization. This multi-scale feature representation enables detecting emotional states by leveraging both local EDA dynamics as well as global topology patterns within the graph-based models [40], and provide relevant information that can be used for identifying emotional states using EDA-graphs [41]. Table II contains the set of 58 features extracted from EDA-graphs, and the literature where the feature was extracted from.

2) Traditional EDA Features: We computed traditional features of EDA for evaluating their suitability for distinguishing emotional states, in comparison with the features obtained from the EDA-graphs. Table III describe the features that we computed such as the mean value of the tonic component of EDA, termed skin conductance level (SCL) [7], the

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TABLE III
FEATURES OF EDA EXTRACTED USING TRADITIONAL METHODS

Feature	Description
SCL	Mean value of the tonic component of EDA
nsSCRs	Number of Non-specific SCRs
TVSymp	The mean value of time-varying spectral amplitudes of EDA between 0.08 and 0.24 Hz
LF (EDASymp)	Sympathetic component of the EDA; Power of the EDA between 0.045 and 0.25 Hz.

number of non-specific skin conductance responses (nsSCRs) [7], the time-varying spectral EDA index of EDA (TVSymp) [21], and the low-frequency sympathetic component of EDA (EDASymp) [22].

E. Statistical Analysis

First, we performed statistical analysis to determine the differences between emotional states for all traditional and EDA-graph features. Then, we use diverse machine learning algorithms to try and classify emotional states using EDA-graph features. We tested the normality of each feature using the Anderson-Darling test [42]. Due to the non-normal distribution of the features, the non-parametric Kruskal-Wallis one way analysis of variance test was initially used to assess the differences between the classes of emotional states for each feature on an individual basis [43]. For each feature yielding a significant Kruskal-Wallis p-value ($p < 0.05$), post-hoc pairwise comparisons between classes were performed using Dunn's test with Holm-Bonferroni correction to control for multiple comparison error. In these post hoc analyses, the Holm-Bonferroni corrected *p-values* from Dunn's tests were used to determine significance levels. This procedure provided a statistically rigorous framework for both assessing overall differences between emotional states through the Kruskal-Wallis test and making inferences about specific emotional state differences through the follow-up corrected Dunn's tests for characteristics that showed significant Kruskal-Wallis results. Significance indicators in Figs. 5 and 6 and are derived from Holm-Bonferroni adjusted *p-values* of post-hoc Dunn's tests. These tests provide insight into specific class-wise differences for features with distributions that were found to be significantly different across emotional state categories in the original Kruskal-Wallis test. In accordance with [44], [45], we used *0.005* as the adjusted significance threshold.

F. Machine Learning Classification

In line with a multiverse analysis approach [46], a diverse range of supervised classifiers were implemented for emotional state classification, including Gaussian Naive Bayes (GNB), KNN, Random Forest (RF), Bagging Classifier (BEC), Adaptive Boosting (Adaboost), Gradient Boosting Classifier (GBC), Decision Trees (DT); and Support Vector Machines (SVM) with Gaussian kernel. This set of machine learning algorithms encompassed probabilistic, instance-based, tree, boosting, and margin-maximizing approaches to enable comparative assessment using methods with different underlying algorithms, inductive biases, and sensitivity to hyperparameters [47]. Grid search was used to extensively tune the hyperparameters of each classifier to optimize their configurations. The different configurations are described in Table IV. The performance of the models was evaluated using leave-one-subject-out cross-validation to account for individual physiological differences. F1 and balanced accuracy were the primary metrics to assess performance in the multi-class emotional states classification.

TABLE IV
HYPERPARAMETERS FOR MACHINE LEARNING OPTIMIZATION

Classifier	Hyperparameters
Naive Bayes (NB)	Variance smoothing: 1e-6, 1e-5, 1e-4
K-Nearest Neighbors (KNN)	Number of neighbors: 1, 2, 3, 4, 5 Weights: uniform, distance Power parameter for the Minkowski metric: 1, 2
Random Forest (RF)	Number of trees: 100, 200 Maximum depth of the tree: 20, 30 Random state: 42
AdaBoost Classifier (ABC)	Number of estimators: 50, 100, 150 Learning rate: 0.1, 0.5
Gradient Boosting Classifier (GBC)	Number of estimators: 50, 100 Learning rate: 0.1, 1.0 Maximum depth of the individual regression estimators: 40, 60
Decision Tree (DT)	Maximum depth: 10, 30 Minimum number of samples required to split an internal node: 2, 10 Random state: 42
Support Vector Machine (SVM)	Penalty parameter C of the error term: 1, 10, 100 Kernel coefficient for 'rbf': 0.1, 1, 10 Kernel type: rbf
Bagging Ensemble Classifier (BEC)	Stacking all the previous classifiers hyperparameters Number of estimators: 50, 100, 500 Max samples: 0.3, 0.5, 1.0 Max features: 0.3, 0.5, 0.5, 0.7, 1 Bootstrap: True, False

G. Feature Importance Analysis

We used the ANOVA F-value feature selection method [48] to rank the features in terms of their power for differentiating between emotional states [13]. This approach systematically ranks features based on their scores derived from an underlying scoring function [49]. (8) defines the scoring function as:

$$score\ (i) = F\left(X_{(i)}, y\right) \tag{8}$$

where $X_{(i)}$ is feature i and y is the target variable. F represents the univariate statistical test of choice. Features are then ranked by their score values and the top k features with the highest scores are selected. This analysis allowed us to get an insight into the graph, node, and edge-level features with the highest emotional states differentiation capabilities [50], [51].

Since it is impractical to plot the values for all 58 features between emotional states, we have included the boxplots of the top 5 EDA-graph features based on ANOVA F-value analysis, shown in Fig. 6.

III. RESULTS

Fig. 3 shows the visual representation of EDA-graphs obtained from EDA segments corresponding to different emotional states (Amused and Relaxed). The EDA-graph corresponding

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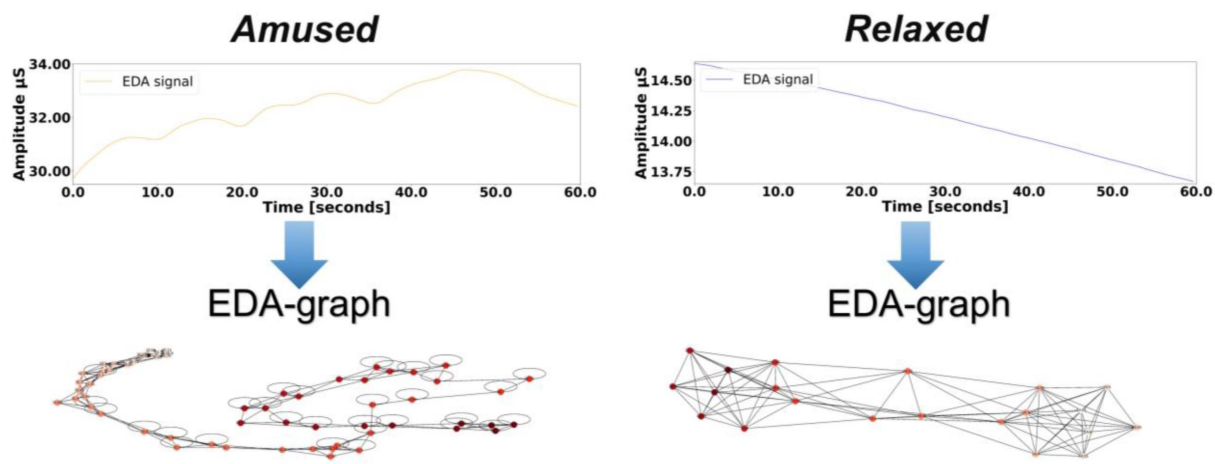

Fig. 3. Samples of EDA signals (top) and EDA-graph representation (bottom) of two EDA segments corresponding to amused (left) and relaxed (right) emotional states.

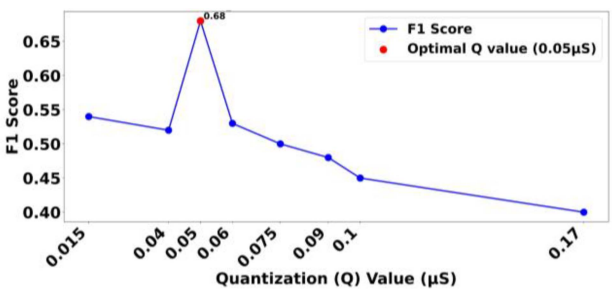

Fig. 4. Selection of optimal Q.

to an Amused segment shows a more irregular pattern, with higher variation in the EDA values compared to the graph of the segment corresponding to a Relaxed segment. As shown in Fig. 4, the Q level of 0.05 μS yielded the highest F1 of 0.68 (SD $= \pm 0.121$), outperforming other Q values that did not exceed 0.55.

Boxplots for EDA features SCL,nsSCRs, TVSymp, and EDASymp showed significant differences between emotional states as illustrated in Fig. 5. SCL was significantly lower in the Bored state compared to Neutral, and Scared states (all $p < 0.005$). SCL did not differ between Bored, Relaxed and Neutral or Amused states ($p < 0.005$). NsSCRs were significantly lower in the Scared state versus all other states and lower in Relaxed versus Amused and Bored and lower in Amused vs Neutral (all $p < 0.005$). NsSCRs did not differ between Relaxed and Neutral (all $p < 0.005$). TVSymp significantly differed between Scared and all other states and between Bored versus Neutral and Amused and between Relaxed versus Bored (all $p < 0.005$). TVSymp was not significantly different between Amused versus Neutral and Relaxed and between Relaxed versus Neutral and Amused (all $p < 0.005$). EDASymp significantly differed between Scared versus Neutral, Amused and Relaxed (all $p < 0.005$). EDASymp did not significantly differ between Amused and Neutral or Relaxed states (all $p < 0.005$).

In Fig. 6 we included the boxplots of the 5 most important EDA-graph features: Total Load Centrality (TLC), Total Harmonic Centrality (THC), Number of Cliques (GNC), Diameter, and Radius. Differences between emotional states are visible in Fig. 6. TLC and THC are node-level features, GNC, Diameter, and Radius are graph-level features. These five features showed significant differences between most or all emotional states pairs ($p < 0.005$). The only exception was GNC, which did not show a significant difference between Amused and Boring.

The machine learning classification results are shown in Table V. The classifiers achieved performance with F1 between 0.50 to 0.56 using traditional features. For EDA-graph methods KNN classifier achieved lower performance with F1 of 0.63. GNB and BEC attained higher F1 of 0.68 in both cases. The GBC, SVM, and DT models exhibited a moderate performance with F1 between 0.66 to 0.65.

The entire collection of features extracted using EDA-graph method, and Traditional EDA methods are included as Python tools and notebooks to enable reproducibility. Features results and tools are now available in the following public GitHub repository https://github.com/jouninLRMD/EDA-graph.

IV. Discussion

We propose a method for obtaining EDA-graphs using GSP to capture complex non-linear associations between EDA signals and emotional states. Additionally, we demonstrate the usefulness of EDA-graphs in detecting emotional states by obtaining sensitive features from the EDA-graphs, which allow for accurate differentiation between emotional states. The graph features showed superior performance compared to traditional EDA features (see asterisk tables in Figs. 5 and 6) [52]. The EDA-graph representation captures nonlinearities while discarding uninformative flat regions. This approach may contribute to inter-state variations by emphasizing informative phasic components.

The multi-scale emotional state-related information within the EDA-graph feature space demonstrates the advantages of graph representations for uncovering complex physiological state relationships relative to conventional signal analysis alone. Our results reveal EDA-graphs' sensitivity to nuanced topological shifts between discrete emotional states, complementing traditional EDA features that are adept at detecting variations in arousal levels [13]. Mapping EDA signals into graph domains unifies these complementary capabilities within a single framework. The ability to generate distinctive graphs for each state validates EDA-graphs' potential for granular differentiation.

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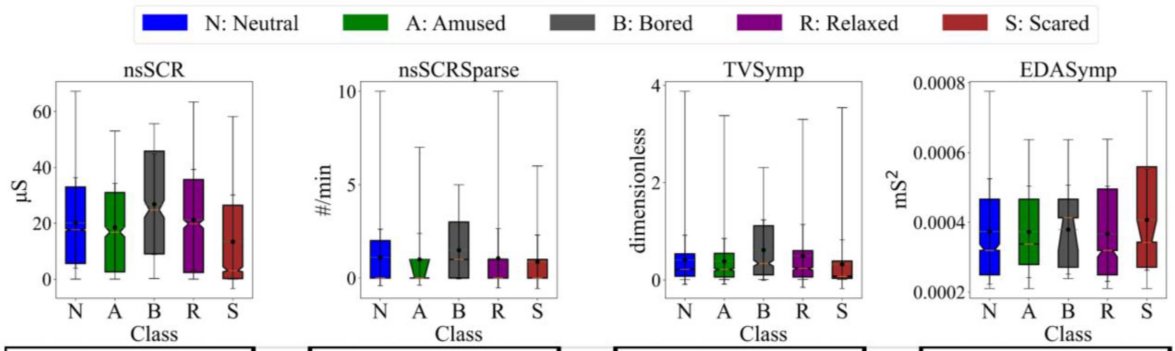


SCL:

	N	A	B	R	S
N			*		
A					
B					*
R					
S					

nsSCR:

	N	A	B	R	S
N		*	*		*
A			*	*	*
B				*	*
R					*
S					

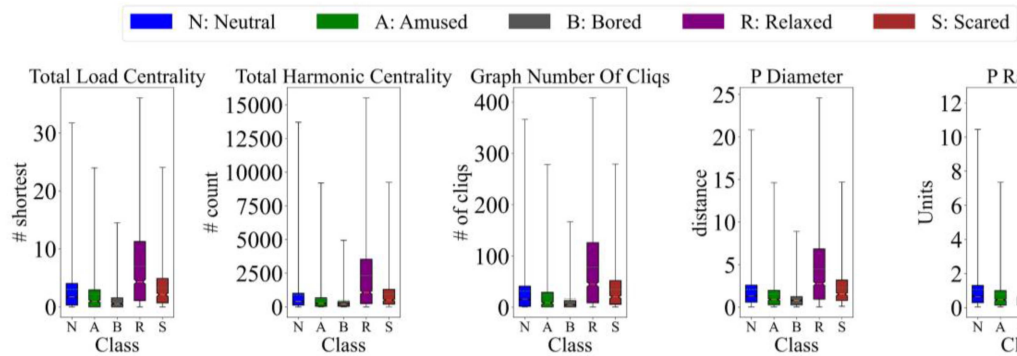
TVSymp:

	N	A	B	R	S
N			*		*
A			*		*
B				*	*
R					*
S					

EDASymp:

	N	A	B	R	S
N					*
A					*
B					
R					*
S					

Fig. 5. Boxplots of the four traditional features of EDA: SCL, nsSCR, TVSymp, EDASymp. The emotional states are labeled as N: Neutral, A: Amused, B: Bored, R: Relaxed, and S: Scared. The tables display with asterisk (∗) the significant differences with $p < 0.005$. The notches in the boxplots represent the confidence interval around the median, whiskers represent the full range of values, lines in the middle represent the mean, and error bars represent standard deviation. The color represents the classes (emotional states).



Total Load Centrality:

	N	A	B	R	S
N		*	*	*	*
A			*	*	*
B				*	*
R					*
S					

Total Harmonic Centrality:

	N	A	B	R	S
N		*	*	*	*
A			*	*	*
B				*	*
R					*
S					

Graph Number of Cliques:

	N	A	B	R	S
N		*	*	*	*
A				*	*
B				*	*
R					*
S					

Diameter:

	N	A	B	R	S
N		*	*	*	*
A			*	*	*
B				*	*
R					*
S					

Radius:

	N	A	B	R	S
N		*	*	*	*
A			*	*	*
B				*	*
R					*
S					

Fig. 6. Boxplots of the five most important features: total load centrality, total harmonic centrality, graph number of cliques, diameter, radius. The emotional states are labeled as N: Neutral, A: Amused, B: Bored, R: Relaxed, and S: Scared. The tables display with asterisk (∗) the significant differences with $p < 0.005$. The notches in the boxplots represent the confidence interval around the median, whiskers represent the full range of values, lines in the middle represent the mean, and error bars represent standard deviation. The color represents the classes (emotional states).

Enhanced characterization of phasic drivers likely also contributes to inter-state differences [53], [54]

Graph-level features, such as total graphlet counts and graph energy, provide a high-level depiction of the global graph structure and complexity. By quantifying the frequency of different graphlet motifs, this metric captures the distribution of small-scale topological structures embedded within the global network. The graphlet profile provides a more granular characterization of network architecture than summary metrics like density or degree distribution [3], [55]. In contrast, node-level features such as centralities and clustering coefficients characterize the local topology and regional signal variations. However,

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TABLE V
CLASSIFICATION PERFORMANCE OF MACHINE LEARNING ALGORITHMS USING TRADITIONAL FEATURES AND EDA-GRAPH FEATURES

	TRADITIONAL METHODS				GRAPH BASED METHODS			
Classifier	**Balanced Accuracy (%)**	**Precision**	**Recall**	**F1**	**Balanced Accuracy (%)**	**Precision**	**Recall**	**F1**
GNB	67.1 ± 10.8	0.66 ± 0.168	0.67 ± 0.108	0.54 ± 0.146	74.7 ± 9.9	0.63 ± 0.157	0.75 ± 0.099	0.68 ± 0.131
KNN	55.7 ± 10.8	0.50 ± 0.151	0.55 ± 0.108	0.52 ± 0.131	66.2 ± 9.3	0.61 ± 0.145	0.66 ± 0.093	0.63 ± 0.127
RF	60.0 ± 11.2	0.49 ± 0.158	0.60 ± 0.112	0.53 ± 0.132	73.4 ± 9.6	0.62 ± 0.141	0.73 ± 0.096	0.66 ± 0.134
Adaboost	67.6 ± 10.3	0.45 ± 0.174	0.67 ± 0.103	0.54 ± 0.144	75.5 ± 10.6	0.61 ± 0.175	0.77 ± 0.106	0.66 ± 0.146
GBC	50.8 ± 9.9	0.49 ± 0.160	0.50 ± 0.099	0.50 ± 0.131	72.6 ± 9.6	0.61 ± 0.143	0.73 ± 0.096	0.66 ± 0.133
DT	56.7 ± 10.5	0.48 ± 0.145	0.56 ± 0.105	0.51 ± 0.142	74.6 ± 10.3	0.62 ± 0.162	0.75 ± 0.103	0.66 ± 0.142
SVM	65.1 ± 9.8	0.50 ± 0.136	0.65 ± 0.098	0.55 ± 0.130	74.9 ± 10.4	0.61 ± 0.145	0.75 ± 0.104	0.66 ± 0.142
BEC	68.4 ± 10.5	0.60 ± 0.162	0.64 ± 0.105	0.56 ± 0.140	74.8 ± 10.1	0.64 ± 0.115	0.74 ± 0.101	0.68 ± 0.121

they lack sensitivity to the overall graph organization. Although degree distributions reveal local connection density, they fail to summarize the entire graph [55].

The purpose of using our EDA-graph method is to accurately capture and interpret the dynamic properties of EDA signals in relation to emotional states. Our approach closely aligns with the inherent dynamics of these signals by applying a threshold for nearest node connections based on the sampling frequency, which is determined through empirical analysis of various EDA samples. Similarly, we have chosen a Q value of 0.05 μS, which has been identified as optimal through comparative analysis. This decision reinforces our commitment to accurately reflecting the nuanced nature of EDA.

We determined the optimal Q value as 0.05 μS through a comprehensive comparative analysis of various quantization levels and their impact on emotion recognition performance. Fig. 4 illustrates the impact of different Q values on the F1 in emotion recognition from EDA signals. This specific Q level allows us to maintain a high degree of accuracy in representing the subtle variations in EDA, crucial for understanding emotional states. A discernible peak at the 0.05 μS mark underscores its optimal performance with an F1 of 0.68, significantly higher than the other Q values, which do not exceed 0.55. The error bars indicate the variability in F1, with the 0.05 μS value having a standard deviation of $\pm$0.121. This graph demonstrates the meticulous selection of 0.05 μS as the preferred Q level for EDA analysis. The method we used can reveal the connectivity between nodes in the EDA-graph context, which represents the temporal interplay of EDA signal points. This allows us to understand how emotional state responses manifest and evolve physiologically. Stronger node connections may indicate more persistent or intense emotional states, while weaker connections might suggest more transient or mild states. Psychologically, this graph-based approach offers an insightful lens into the dynamics of emotional responses, revealing patterns and relationships that might be overlooked in traditional analysis methods.

The EDA-graph features were more effective than the traditional EDA features in distinguishing between emotional states. The nsSCR feature was the most effective traditional feature for differentiating emotional states. However, both nsSCR and TVSymp primarily identified differences between the Scared state and other states. Notably, except when compared to the Relaxed state, neither nsSCR nor TVSymp showed significant differences for the Amused state. These features have limitations in detecting nuanced differences related to amusement. EDASymp and SCL were not effective in distinguishing between high and low arousal states, indicating low sensitivity to changes in arousal or valence.

The lower SCL for Neutral, Arousal and Relaxed aligns with decreased sympathetic arousal during relaxation, consistent with SCL's known arousal detection capabilities [56]. Meanwhile, traditional features provided some valuable insights, they exhibited limitations in making nuanced distinctions between emotional states that the EDA-graph features were better able to detect. This further motivates developing nonlinear EDA analysis methods like EDA-graphs for optimized emotional states recognition from EDA data.

The graph representation of EDA signals provides valuable insights into the explainability of the observed differences between emotional states. The node-level features, such as TLC and THC capture nuanced variations in local connectivity patterns and information propagation efficiency within the EDA-graphs. For instance, higher TLC suggests the presence of "hub" nodes that facilitate connectivity across the network, which could be indicative of concentrated emotional responses with focal points that integrate activity [55]. The highly significant *p-values* for TLC differences between neutral and states like amused (p = 1.87e-17), bored (p = *2.71e-26*), relaxed (p = 4.83e-108), and scared (p = 5.53e-08) support this interpretation. Similarly, higher THC implies more efficient signal propagation via short walks between nodes, possibly associated with distributed emotional responses. The *p-values* reveal highly significant THC differences between neutral and other states (amused p = 1.69e-19, bored p = 2.26e-22, relaxed p = 1.45e-98, scared p = 7.31e-09). In addition, graph-level features like GNC, Diameter, and Radius highlight distinctions in global structure, community organization, reachability, and network compactness associated with different affective states. Emotional states eliciting larger diameters (e.g., neutral vs. amused p = 3.12e-19, bored p = 9.22e-23, relaxed p = 7.28e-100, scared p = 1.20e-08) may reflect sparser, less integrated network topologies compared to those with smaller diameters. This graph-based characterization provides an interpretable window into how physiological signals manifest intricate patterns associated with discrete emotional experiences, potentially revealing insights overlooked by conventional signal analysis techniques that focus solely on amplitude variations.

TLC calculates the number of shortest paths between all pairs of nodes in the network that pass through each given node. The shortest path represents the minimum number of edges that must be traversed to connect two nodes. Nodes with higher load centrality are more "central" because more of the shortest paths in the overall network go through them [75]. This indicates the node is more important for facilitating connectivity across the graph. In network science, these nodes with high centrality are called "hubs" [60]. Hubs act as focal points that integrate and allow flow across the network. Differences in TLC

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between emotional states graphs suggest some affective states lead to more concentrated connectivity where fewer hub nodes integrate into the network. Other states exhibit more distributed connectivity with shortest paths spread across many nodes rather than through a few hubs.

THC is the harmonic mean of the number of closed walks starting from each node. A closed walk is a path that starts and ends at the same node [79]. The harmonic mean accounts for both short walks involving few edges and long walks involving more edges. Higher harmonic centrality indicates a node can reach other nodes more quickly through short walks, implying high network integration [64].

Differences in THC suggest some emotional states graphs have topology that enables faster information propagation across the network compared to others. Community structure refers to the organization of nodes into tightly interconnected groups. Efficiency describes how easily signals flow through the network. Small-world characteristics include high local clustering of connections and short paths between distant nodes.

This likely reflects differences in community structure, efficiency of connections, and small-world characteristics that impact how easily signals can travel in the network. Graphs with well-connected communities, efficient connections, and small-world properties tend to have higher THC as signals spread rapidly via short walks. Lower THC indicates networks where signals propagate less efficiently due to topological differences [65], [66].

GNC counts the total number of cliques in the graph. A clique is a fully connected subgraph where each node is connected to all other nodes. Cliques represent sets of nodes that are highly interconnected with each other, forming cohesive subgroups. GNC quantifies how many of these densely interconnected communities exist [67], [68]. The lack of difference in GNC between amused and boring graphs suggests they may have similar local cohesiveness and modular structure despite differences in global topology. Additional analysis of the local substructures could provide insights into differences in interconnectivity patterns elicited by specific emotional states [86].

Diameter represents the longest shortest path between any pair of nodes in the network [70]. Radius calculates the minimum eccentricity, which is the maximum shortest path length between a node and all other nodes [71]. These compactness measures showed significant differences between all emotional states' graphs. Some states elicit longer paths and greater dispersion, indicating lower connectivity and efficiency. Other states have shorter diameter and radius, suggesting more clustered and integrated topology with dense local connectivity. This reveals how emotional states induce differential patterns of reachability, sparseness, and spread in the networks [90].

Graph metrics such as clustering coefficients and centrality provide deeper insights. High clustering in the EDA graphs suggests concentrated EDA activity, indicative of focused or intense emotional responses. In contrast, nodes with high centrality, acting as hubs, can be seen as influential states that play a pivotal role in governing the overall emotional response pattern [55]. This granular analysis, enabled by GSP, offers a comprehensive understanding of the complex interplay between physiological signals and emotional states, potentially revealing patterns that conventional analysis methods might overlook.

Comparative analysis showed that conventional EDA features can detect changes but have limitations in characterizing nuanced emotional states. In contrast, graph features demonstrated greater sensitivity to subtle state variations, aligning with evidence on complementarity of physiological signals and network representations [55]. This pioneering study introduced an EDA-graph framework that captures complex nonlinear emotional state-signal relationships and enables granular state differentiation. The results represent a significant advance toward graph-based emotional states recognition systems using physiological signals. Further research should focus on evaluating this approach across expanded affective categories and combining it with ensemble classifiers [69].

The machine learning classification performance using traditional EDA features Table V showed limited accuracy, with maximum F1 between 0.56 and 0.56. In contrast, the classifiers achieved significantly improved predictive ability when utilizing the proposed EDA-graph features Table V. Classifiers including GNB, KNN, RF, AdaBoost, GBC, DT, SVM, and BEC attained F1 between 0.63 and 0.68, using the graph representations, demonstrating their usefulness for learning complex topological patterns associated with discrete emotional states. The ensemble methods showed strong results, converging on the emotional states-differentiating information encoded within the EDA graph topology metrics.

GSP methods are used in affective computing and Human-Computer Interaction (HCI) to analyze emotional and physiological datasets. Chanel and Mühl (2015) discuss how integrating GSP can enhance HCI by fostering social connectivity [75]. GSP is ideal for emotional analysis as it can interpret relational and dynamic data. Liu et al. (2021) demonstrate the application of GSP in facial affect analysis, improving emotion recognition in HCI [76]. GSP captures the nuanced dynamics of emotional states and is crucial for real-time analysis in affective computing. Gui et al. (2019) highlight the need for rapid graph processing in such scenarios [77]. Wang et al. (2022) propose that incorporating GSP with multimodal data enhances emotion recognition systems [78]. Nodes and edges within graphs represent emotional and physiological signals, facilitating the understanding of complex patterns and structures in affective data. This aligns with the goal of improving emotional state detection and interpretation in HCI applications.

State-of-the-art methods exhibit certain features that differentiate between neutral and other emotional states. netEDA presents an Assortativity feature that provides substantial differentiation between neutral and other emotional states, as evidenced by low *p-values*, especially between neutral and bored. comEDA's Quadratic Rényi entropy feature demonstrates significant discrimination between neutral, amused, and bored emotions, denoting its potential in enhancing emotion classification. topEDA's Area feature stands out for its discriminative performance, particularly between neutral and bored emotions, emphasizing its effectiveness in categorizing emotional states. Finally, isaxEDA's Shannon entropy feature, derived from skewness-based Piecewise aggregate approximation (PAA), showcases robust discrimination between neutral, amused, and bored emotions, reinforcing its prominence in differentiating emotional states [79]. Compared to these established methods, our EDA-graph approach not only demonstrates superior performance in emotional state recognition, but also illustrates the intricate interplay of EDA-graph features in capturing emotional subtleties. The node-level and graph-level features of the EDA-graph work together to provide a comprehensive representation of emotional states.

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TABLE VI
COMPARATIVE ANALYSIS OF EMOTIONAL STATES RECOGNITION METHODS USING THE CASE DATASET

Method	Accuracy (%)	F1
Few-shot Learning (Zhang et al.)	58.77	0.482
netEDA (Random Forest)	70.00 ± 11.7	0.640 ± 0.123
comEDA (Naïve Bayes)	67.00 ± 9.6	0.550 ± 0.108
topEDA (Random Forest)	69.00 ± 9.8	0.630 ± 0.099
isaxEDA	69.00 ± 9.2	0.650 ± 0.104
Traditional EDA Features (BEC)	68.40 ± 10.5	0.560 ± 0.140
EDA-Graph Features	75.50 ± 10.6	0.685 ± 0.131

TABLE VII
RUNNING TIMES KEY FINDINGS

Method	Time for calculating features per sample (seconds)	Total time for feature extraction (seconds)	Time for machine learning classification (seconds)
Traditional EDA Feature Extraction	0.012	197 (~3 minutes)	1182 (~19.7 minutes or ~0.3 hours)
EDA-graph Feature Extraction	0.025	411 (~6.8 minutes)	1644 (~27.4 minutes or ~0.45 hours)
Few-shot Learning (5-shot-5C)	-	-	419.25
Few-shot Learning (10-shot-5C)	-	-	1543.25

Our comparative analysis, detailed in Table VI included an analysis of Zhang et al.'s few-shot learning method, which achieved 58.7% accuracy and an F1 of 0.48 [26]. We also evaluated four advanced non-linear signal processing techniques by Veeranki et al. isaxEDA, which attained an accuracy 69.0% accuracy and 0.65 F1; comEDA, with 67.0% accuracy and 0.55 F1; topEDA, reaching 69.0% accuracy and 0.63 F1; and netEDA, with 70.0% accuracy and 0.64 F1 [79]. Additionally, employing traditional EDA features through the BEC classifier yielded an accuracy of 68.40% and an F1 of 0.56. Our research introduces the EDA-graph method, which significantly outperformed the approaches with a remarkable accuracy of 75.5% and an F1 of 0.68. These comparative results, as depicted in Table VI, this not only demonstrates the superior performance of our method in recognizing emotional states but also marks a substantial improvement over both the conventional EDA feature-based approaches and the innovative few-shot learning method utilized by Zhang et al. The running time analysis further supports the EDA-graph's efficiency, showing a balance between computational demand and performance, thereby endorsing its applicability in real-world scenarios as presented in Table VII. These results underscore the effectiveness of the EDA-graph method as a significant contribution to the field of emotion recognition. Suggesting that our approach provides a more accurate and reliable means of analyzing emotional states from EDA signals, outpacing current methodologies including few-shot learning, which is notable for its requirement of minimal data.

The comparative outcomes highlight the advantages of graph-based learning approaches over traditional EDA features for unlocking affective signals within the physiological data. The EDA-graph methodology provides a novel means of transforming EDA data into a domain that enhances emotional states characterization. Ongoing research can build on these promising results by further optimizing graph architectures and machine learning pipelines tailored for emotion recognition.

The proposed EDA-graph approach has limitations: sample size too small for generalizability, analysis focused on limited emotional states, and reliance on only EDA. Expanding the analysis to include a wider range of emotional states and integrating multiple modalities could enhance accuracy. More systematic exploration of graph construction parameters and consideration of individual differences are needed. Collaborative efforts across disciplines are crucial for a deeper understanding of physiological signals, graph representations, and emotional experiences.

For future work, we aim to expand the EDA-graph framework by integrating Graph Neural Network (GNN) methodologies to enhance pattern recognition. This progression will focus on automated feature extraction, potentially leveraging the Graph Attention Mechanism and Adaptive Graph Convolutional models, such as advancements in sleep network analysis, to refine our emotion classification approach directly from the graph structure [80], [81].

V. CONCLUSION

This study presented a novel GSP methodology for EDA analysis using EDA-graphs, which represent EDA signals by linking signal values based on Euclidean distance. Statistical analysis revealed EDA-graph node- and graph-level features, including TLC, THC, GNC, Diameter, and Radius, exhibit significant differences between emotional states. This demonstrates the ability to capture topological patterns in EDA associated with affective shifts. Using the graph features, machine learning classifiers achieved maximum cross-validation F1 up to 0.68, surpassing performance with traditional EDA features. While accuracy is still improvable, these promising results provide initial validation of modeling EDA as graph representations to unlock signals relevant for emotion detection. Ongoing work should focus on optimizing graph architectures and machine learning pipelines to further advance emotion recognition performance. Overall, this pioneering study introduced a graph-based framework for eliciting emotion encoding within EDA data, motivating continued research into specialized graph learning approaches for affective state characterization.

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